Chapter 1

Vector Space

A vector space V over a field $\mathbb{F}$ is a set with two binary operations: scalar multiplication and vector addition. For vectors $\vec{v}, \vec{u}$, and $\vec{w}$, and any scalars $a, b \in \mathbb{F}$:

1. $\vec{u} + \vec{v} = \vec{v} + \vec{u}$
2. $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$. Repeated applications of this axiom allow us to write the sum of n vectors without parenthesis needed.
3. There exists $\vec{0} \in V$ such that $\vec{v} + \vec{0} = \vec{v}$. The vector $\vec{0}$ is unique in V
4. For all $\vec{v} \in V$ there exists $\vec{v}' \in V$ such that $\vec{v} + \vec{v}' = \vec{0}$. The vector $\vec{v}'$ is unique and may be denoted $-\vec{v}$.
5. For $1 \in \mathbb{F}$, $1\vec{u} = \vec{u}$
6. $(ab)\vec{u} = a(b\vec{u})$
7. $a(\vec{u} + \vec{v}) = a\vec{u} + a\vec{v}$
8. $(a + b)\vec{u} = a\vec{u} + b\vec{u}$

Example 0.0.1: Examples of Vector Spaces

Fundamental vector spaces include:

1. $\mathbb{F}^n$: the set of n-tuples in $\mathbb{F}$ with coordinate wise vector addtiion and scalar multiplication (think $\mathbb{R}, \mathbb{R}^2, \mathbb{R}^n$ and $\mathbb{C}^n$)
2. $P(\mathbb{F})$: the set of *all* polynomials with coefficents in $\mathbb{F}$
3. $P_n(\mathbb{F})$: the set of all degree n polynomials with coefficents in $\mathbb{F}$
4. $\mathbb{M}_{n\times m}(\mathbb{F})$: the set of all $m \times n$ matricies with entries in $\mathbb{F}$ with vector addition and scalar multiplication defined as usual
5. $\mathcal{F}(S, \mathbb{F})$: the set of all functions $f : S \to \mathbb{F}$ for $S \neq \varnothing$

Subspaces

Let V be a vector space over $\mathbb{F}$. Then $W \subseteq V$ is a subspace of V if, and only if, W is a vector space over $\mathbb{F}$ with scalar multiplication and vector addition defined as in V. Noteably, V and $\{0\}$ are subspaces of V.

Given $W \subseteq V$ to determind if W is a subspace of V it suffices to check that

1. $0 \in W$
2. $\vec{u} + \vec{v} \in W$ for $\vec{u}, \vec{v} \in W$
3. $a\vec{u} \in W$ for $a \in \mathbb{F}$

Elementary Row Operations & RREF

Let A be an $m \times n$ matrix. Given any matrix A we may apply the following elementary row operations:

- $r_i \to r_i - nr_j$ for $n \in \mathbb{F}$
- $r_i \leftrightarrow r_j$
- $r_i \to nr_i$ for $n \in \mathbb{F}$

A matrix A is said to be in RREF (reduced row echelon form) if all its pivots (first non-zero entry in a row) are 1. We may apply elementary row operations to A to transform it into A_{RREF}. In a system of equations we use elementary row operations to go from $(A \mid b) \to (R \mid c)$ which preserves solutions to the system.

Linear Combination & Span

If V is a vector space and $S \subseteq V \neq \varnothing$ then $\vec{v} \in V$ is a linear combination of vectors in S if

$$\vec{v} = \sum_{i=0}^{n} a_i \vec{u}_i$$

where $a_i \in \mathbb{F}$ and $\vec{u}_i \in S$ for all $i = 1, \ldots, n$ (S must be finite). The scalars $a_1, \ldots, a_n$ are called the coefficents of the linear combination. The span of S is the set of all linear combinations of S. Specifically we say that $\text{Span}\{\varnothing\} = \{0\}$. If $\text{Span}\{S\} = V$ for a vector space V, we say that S generates V.

If S is a set, we can determine if a vector $\vec{v}$ is in its span by row reducing $(A \mid \vec{v})$ where A is the matrix formed by writing all the vectors in S as columns vectors. If the system is consistent (i.e. no pivots in the last column) then $\vec{v} \in \text{Span}\{S\}$ and otherwise $\vec{v} \notin \text{Span}\{S\}$.

If S is a set, we can determine if $\text{Span}\{S\} = \mathbb{F}^n$ by row reducing the matrix A that is formed by writing the vectors of S as column vectors and checking if $\text{rank}(A) = n$ (recall that the rank of a matrix is equal to the number of pivots when row reduced).

Linear Independence/Dependence

Consider $S \subseteq V$ for some vector space V. The set S is said to be linearly dependent if there exist a finite amount of vectors $\vec{u}_1, \ldots, \vec{u}_n \in S$ and scalars $a_1, \ldots, a_n \in \mathbb{F}$ not all zero such that

$$\sum_{i=0}^{n} a_i \vec{u}_i = 0.$$

When $a_1 = a_2 = \cdots = a_n = 0$, we call it the trivial representation of 0. If the trivial representation of 0 is the only representation of 0 then S is said to be linearly independent.

Given a set of vectors S we may check if it linearly independent or dependent by looking at the augmented matrix $(A \mid 0)$ (which we can treat as simply just A), where A is the matrix formed by writing the vectors in S as column vectors. If there are infinitely many solutions (i.e. if there is at least one free variable) then S is linearly dependent. If there is only a single solution then S is linearly dependent.

Basis

A basis β for a vector space V is a linearly independent subset of V that generates V.

Theorem 0.1. *Let V be a vector space and $u_1, \dots, u_n$ be distinct vectors in V. Then $\beta = \{u_1, \dots, u_n\}$ is a basis for V if, and only if, each $v \in V$ can uniquely be expressed as a linear combination of vectors of β, that is, can be expressed in the form*

$$v = a_1 u_1 + \dots a_n u_n$$

for unique scalars $a_1, \dots, a_n \in \mathbb{F}$.

Proof. ($\Rightarrow$) Let β be a basis for V. It follows that for any $v \in V$, $v \in \operatorname{Span}\{\beta\}$ since $\operatorname{Span}\{\beta\} = V$. We now show that the representation of v is unique. Consider two representations of v:

$$v = a_1 u_1 + a_2 u_2 + \dots + a_n u_n$$
$$v = b_1 u_1 + b_2 u_2 + \dots + b_n u_n$$

however, if we subtract the two then we see that

$$0 = u_1(a_1 - b_1) + \dots + u_n(a_n - b_n)$$

since $u_i \neq 0$ it follows that $a_1 - b_1 = 0 \Rightarrow a_1 = b_1$ and thus the representation of v is unique.
($\Leftarrow$) Conversely, assume that each $v \in V$ can be uniquely expressed as a linear combination of vectors in β. It clearly follows that β is a basis for V since, $\operatorname{Span}\{\beta\} = V$ because every vector in V can be represented as a linear combination of vectors in β by assumption. As well, since representations are unique it must be true that β is linearly independent because there is a unique representation of $0 \in V$. □

Dimension

A vector space V is said to be finite-dimensional if it has at least one basis consiting of a finite number of vectors, otherwise V is called infinite-dimensional. The dimension of a vector space is denoted $\dim V$. Specifically, we say that $\dim\{0\} = 0$ and $\dim \mathbb{F}^n = n$.

Following from the Replacement Theorem (Thm1.10 in the textbook) we have that for a vector space V that has a finite basis, it follows that *all* bases for V are also finite and contain the same number of vectors. As well, if $\dim V = n$ then

1. Any finite generating set for V contains at least n vectors, and a generating set of V that contains exactly n vectors is a basis for V
2. Any linearly independent subset of V that contains exactly n vectors is a basis for V
3. Every linearly independent subset of V can be extended to a basis for V, that is, if L is a linearly independent subset of V, then there exists a basis β such that $L \subseteq \beta$.

Example 0.0.2: Standard Basis

- For $\mathbb{R}^n$ we denote the standard basis to be

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

  so that for e_i the i-th row is 1 and every other row is 0.

- For $P_n(\mathbb{F})$ we denote the standard basis to be

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ x \\ \vdots \\ 0 \end{pmatrix}, \dots e_{n+1} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ x^n \end{pmatrix}$$

so that for e_i the i-th row is x^{i-1} and every other row is 0.

Chapter 2

Linear Transformations

Let V and W be vector spaces over $\mathbb{F}$ and define $T : V \to W$. T is a linear transformation if for all $x, y \in V$ and $c \in \mathbb{F}$ we have

1. $T(x + y) = T(x) + T(y)$
2. $T(cx) = cT(x)$

If T is linear it follows that $T(0) = 0$ and that $T(x - y) = T(x) - T(y)$.

Null Space & Range

Let V and W be vector spaces and $T : V \to W$ be a linear transformation. The null space of T is defined to be the set of all vectors x such that $T(x) = 0$. Symbolically

$$N(T) = \{x \mid T(x) = 0\}.$$

The nullity of T is the dimension of $N(T)$. The range of T is the set of all images of vectors x:

$$R(T) = \{T(x) \mid x \in V\}.$$

The rank of G is the dimension of $R(T)$.

Theorem 0.2 (Dimension Theorem)**.** *Let V and W be vector spaces and $T : V \to W$ be linear. If V is finite dimensional then*

$$nul\,(T) + rank\,(T) = \dim V$$

Theorem 0.3. *Let V and W be vector spaces and $T : V \to W$ be linear. Then T is one-to-one if, and only if, $N(T) = \{0\}$*

Proof. ($\Rightarrow$) Assume T is one-to-one. It follows by definition of injectivity that $N(T) = \{0\}$ since if $T(x) = T(y) = 0$ then $x = y = 0$ since $T(0) = 0$ if T is linear.
($\Leftarrow$) Conversely, assume that $N(T) = \{0\}$. We must show that $T(x) = T(y)$ implies that $x = y$. However, if $T(x) = T(y)$ then

$$0 = T(x) - T(y) = T(x - y)$$

thus $x - y \in N(T) = \{0\}$ so $x - y = 0 \Rightarrow x = y$ and so T is injective. □

Theorem 0.4. *Let V and W be finite dimensional vector spaces and $T : V \to W$ be linear. Then the following are equivalent:*

1. *T is one-to-one*
2. *T is onto*
3. *$rank(T) = \dim V$*

Theorem 0.5. *Let V and W be vector spaces over $\mathbb{F}$ and suppose that $v_1, v_2, \ldots, v_n$ is a basis for V. For $w_1, w_2, \ldots, w_n$ in W there exists exactly one linear transformation $T : V \to W$ such that $T(v_i) = w_i$*

Proof. Let $x \in V$ so that

$$x = \sum_{i=1}^{n} a_i v_i$$

for unique $a_i \in \mathbb{F}$. Then define $T : V \to W$ by

$$T(x) = \sum_{i=1}^{n} a_i w_i$$

so that clearly $T(v_i) = w_i$. To show T is linear consider $x, y \in V$ and $c \in \mathbb{F}$. Then

$$T(x+y) = \sum_{i=1}^{n} (a_i + b_i) w_i = \sum_{i=1}^{n} a_i w_i + \sum_{i=1}^{n} b_i w_i = T(x) + T(y)$$

where

$$x = \sum_{i=1}^{n} a_i v_i \quad \text{and} \quad y = \sum_{i=1}^{n} b_i v_i$$

for unique scalars $a_i, b_i \in \mathbb{F}$ for $i = 1, 2, \ldots, n$. Also,

$$T(cx) = \sum_{i=1}^{n} a_i c w_i = c \sum_{i=1}^{n} a_i w_i = cT(x).$$

Now, to show uniqueness consider some $U : V \to W$ such that $U(v_i) = w_i$ as well. So

$$U(x) = \sum_{i=1}^{n} a_i U(v_i) = \sum_{i=1}^{n} a_i w_i = T(x)$$

and hence $U = T$ so T is unique. □

Ordered Basis

Let V be a finite dimensional vector space with basis β. Then, β is an ordered basis if it is given a specific order

Coordinate Vectors

Let $\beta = \{v_1, \ldots, v_n\}$ be an ordered basis for a finite dimensional vector field V. If

$$x = \sum_{i=1}^{n} a_i v_i$$

for unique scalars $a_i \in \mathbb{F}$ for $i = 1, \ldots, n$ then the coordinate vector of x relative to β is

$$[x]_\beta = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

Matrix Representation of T

Let β and γ be ordered bases for vector spaces V and W respectively, and consider $T : V \to W$. Then, the $m \times n$ matrix A comprised of the column vectors $[T(v_i)]_\gamma$ is called the matrix representation of T in the ordered bases β and γ. We say $A = [T]_\beta^\gamma$ (or $A = [T]_\beta$ if $V = W$).